

Lab 3 – Architecture Justification

Virtual Technical Conference Platform | Nehaa Joshi | PES1UG24CS297

1. Architecture Selected: Layered Architecture

The platform is divided into three layers: the **Presentation Layer** (Conference Web UI, Conference Admin UI), the **Business Layer** (Authentication & Access Control, Paper Review Manager, Live Q&A Manager, Schedule Manager) and the **Data Layer** (Conference Database). Each layer only uses the layer directly below it, through well-defined interfaces.

2. Why this architecture suits the problem

Reason 1 – Separation of concerns. The system has three distinct responsibilities: double-blind paper review, conflict-free multi-track scheduling, and real-time upvoted Q&A. Giving each its own business component keeps their rules (anonymisation, room/speaker constraints, vote ordering) independent of each other and of the user interface.

Reason 2 – Maintainability and modifiability. Because components talk only through interfaces, the UI can be redesigned or the database replaced without changing business logic, and each manager can be developed and tested separately.

3. Security Advantage

Authentication & Access Control is a single, isolated component that every manager depends on through the *Role & Access Control Service* interface, so permissions for authors, reviewers, organisers and attendees are enforced in one place. The UI layer has no direct path to the database, and the Paper Review Manager releases only anonymised manuscripts to reviewers, which protects the double-blind requirement.

4. Performance Benefit

Business functions are independent, so the heavily used Live Q&A Manager can be optimised or scaled (for example with caching of vote counts) without affecting paper review or scheduling. All database access goes through the single *Database Query Interface*, which allows centralised connection pooling and query optimisation. The trade-off is one extra hop between layers, which is small compared with the maintainability and security gains.

5. Components and Interfaces (summary of diagram)

Interface	Provided by	Required by
User Authentication API	Authentication & Access Control	Conference Web UI
Paper Submission / Review API	Paper Review Manager	Conference Web UI
Real-Time Q&A API	Live Q&A Manager	Conference Web UI
Schedule Management API	Schedule Manager	Conference Admin UI
Role & Access Control Service	Authentication & Access Control	Paper Review, Schedule, Live Q&A Managers
Database Query Interface	Conference Database	All four business components